

Syed Arbaaz Qureshi

+1 (413) 695-7408 | arbaaz.qureshi29@gmail.com | linkedin.com/in/arbaazqureshi | github.com/arbaazQureshi | arbaazqureshi.me

SUMMARY

Applied scientist with **5+ years of experience** at **Google, Amazon, Microsoft, IBM, and Lowe's**, in **LLMs, multimodal agents, conversational AI, search, recommendation, and personalization**. Delivered **multi-million dollar revenue** through **scalable, maintainable, and robust** solutions in large **cross-domain teams**. Proven **research** record with **8 top-tier publications (AAAI, IJCNN)**, cited **240+ times** with an **h-index of 4**.

EDUCATION

University of Massachusetts Amherst

Master of Science, Computer Science, GPA: 4.0/4.0

Amherst, MA

Sep 2021 – May 2023

Relevant courses: Machine Learning, Advanced NLP, Applied IR, Algorithms for Data Science, Distributed Systems

Indian Institute of Technology Patna

Bachelor of Technology, Computer Science and Engineering, GPA: 8.2/10

Patna, India

Aug 2015 – May 2019

Relevant courses: Algorithms and Data Structures, Foundations of ML, AI, NLP, Introduction to Deep Learning

EXPERIENCE

Amazon

December 2024 – Current

Applied Scientist | Manager: Dhruv Agarwal | Media coverage: Forbes, The Verge

Seattle, WA

- Built and deployed **LLM-powered web agents** that accurately pick and confirm user-specified product attributes on product pages, as a part of the **core science team** that **automates shopping** on third-party (non-Amazon) e-commerce sites.
- Developed step-level **verifiers** that validate agent state against user intent and gate irreversible steps (add-to-cart, order placement), detecting deviations and triggering **recovery** by providing corrective **feedback**, cutting incorrect purchases by **93%**.

Lowe's

July 2023 – December 2024

Applied Scientist | Manager: Dr. Surya Kallumadi

Charlotte, NC

- Developed an agentic multimodal **context-aware conversational recommender system** using **LLMs**, which facilitates product discovery and comparison, and answers home improvement queries. This work is **demoed to and closely followed by the CTO** of Lowe's, Seemantini Godbole.
- Engineered a **personalized** session-aware **retriever** and **re-ranker** for the **query auto-complete** system using open-source **text embeddings (transformers)** and ANN indexes (**HNSW, ScaNN**), resulting in **2.2% engagement increase**, and **\$37M annual incremental revenue**.

IBM

Feb 2023 – June 2023

Graduate Student Researcher | Managers: Dr. Taesung Lee and Dr. Youngja Park

Remote, USA

- Developed a method for **generating text descriptions** for concept-activated neurons in **BERT** without human intervention.
- Created and analyzed a 72K Amazon review dataset with annotated concepts using **open-source LLMs**, mapping trigger concept to neurons. Findings **published in AAAI 2024 (ReLM)**.

Google

Sep 2022 – Dec 2022

Software Engineering Intern | Manager: Dr. Cac Nguyen

Mountain View, CA

- Developed a **low-latency off-body detection** system with **convolutional neural networks** for the **Google Pixel Watch**, including an end-to-end framework for data processing, network training, and real-world evaluation.
- Achieved **competitive performance**, matching production standards for **latency ($\leq 1s$)**, **sensitivity ($\geq 95\%$)**, and **false trigger rate (≤ 1 per day)**.

Lowe's

May 2022 – Aug 2022

Graduate Applied Science Intern | Manager: Dr. Surya Kallumadi

Charlotte, NC

- Developed and deployed **eXtreme Multi-label Classification (XMC)** models for **semantic product search** and **query auto-completion** using **PECOS**, achieving competitive recall with existing systems and **improving per-query latency by over 20x**.
- Successfully integrated XMC models into production, now serving approximately **20 million customer transactions weekly**, resulting in **2.5% conversion rate improvement** and creating an **annualized incremental value of over \$22M**.

Microsoft Research

Aug 2019 – Aug 2021

Research Engineer | Managers: Dr. Sonu Mehta, Dr. Rahul Kumar & Dr. Ranjita Bhagwan

Bangalore, India

- Developed a code diff to natural language model for **automatically generating commit messages**, and **edit2vec** for **code edit classification**. Trained both models on **datasets** scraped from **over 1000 GitHub repositories**, and achieved over **99% accuracy in classifying code edits**.
- Identified limitations of code2vec** in generalizability to other downstream tasks during the development and evaluation process.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C#, MySQL

Toolkits: PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, Docker, LangChain, LangGraph, DSPy, Google Cloud Platform, Unix/Linux

Core Areas (computer science): Reinforcement Learning, Machine Learning, Deep Learning, Natural Language Processing, Information Retrieval, Recommender Systems, Data Structures & Algorithms, Software Engineering

Core Areas (mathematics): Probability & Statistics, Linear Algebra, Multivariate Calculus, Convex & Non-Convex Optimization, Numerical Methods

SELECTED PROJECTS

Research Paper Tagger (RPT)

Sep 2021 – Dec 2021

Advisor: Dr. Mohit Iyer | Code & paper: RPT GitHub repo

UMass Amherst

- Built and **fine-tuned** various **BERT-based classifiers** on a dataset of 1,744 ACL research papers to **automatically tag research tracks** using titles, abstracts, and author information.
- Achieved top-1 accuracy of 70% and top-3 accuracy of 83% on the test dataset..

Multimodal and multi-task representation learning for estimating depression severity

Aug 2018 – May 2019

Advisors: Dr. Sriparna Saha, Dr. Gaël Dias & Dr. Mohammed Hasanuzzaman

AI-NLP-ML lab, IIT Patna

- Developed and trained **multimodal learning** networks for **depression severity estimation**, achieving **4.93% improvement in RMSE** and **1.50% in MAE**, **beating the SoTA** with **66.66% accuracy** and **0.53 F1-score** on the **DAIC-WoZ dataset**.
- Designed and trained various **multitask learning** models to predict both **emotion intensity** and **depression severity** from text data, demonstrating significant **performance gains with emotion and gender-aware** approaches. Findings **published in IJCNN, IEEE CIM, IEEE IS**.

Cycle Location and Anti-Theft System (CLATS)

Jan 2017 – May 2017

Advisor: Dr. Jimson Mathew

IIT Patna

- Developed a lightweight **RFID-based system** and **Android app** to live-track bicycles and prevent unauthorized use, featuring a digital lock with alarm functionality.
- Awarded "**Most Innovative Project**" and selected for a **patent** by the Computer Science faculty at IIT Patna.

SELECTED PUBLICATIONS

1. **Towards generating informative textual description for neurons in language models**
Association for the Advancement of Artificial Intelligence (AAAI), 2024 [pdf]
2. **Gender-Aware Estimation of Depression Severity Level in a Multimodal Setting**
International Joint Conference on Neural Networks (IJCNN), 2021 [pdf]
3. **Improving Depression Level Estimation by Concurrently Learning Emotion Intensity**
IEEE Computational Intelligence Magazine (IEEE CIM), Volume 15, Issue 3, 2020. [pdf]
4. **Multitask Representation Learning for Multimodal Estimation of Depression Level**
IEEE Intelligent Systems (IEEE IS), Volume 34, Issue 5, 2019. [pdf]
5. **Automatic Prediction of PHQ-8 Questionnaire Scores using Artificial Intelligence**
French Journal of Psychiatry, Volume 1, Supplement 2, 2019 [pdf]

MEDIA COVERAGE

1. **Amazon 'Buy For Me' Is The Latest Entrant In The AI Shopping Agent Race**
Forbes, April 2025. [link]